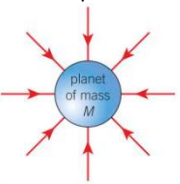
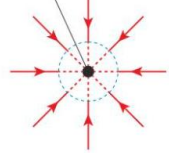
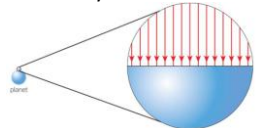
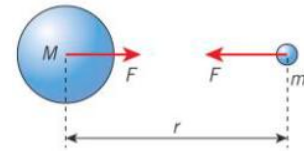
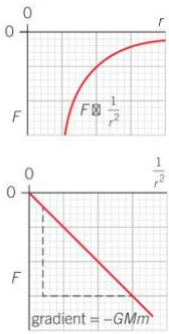


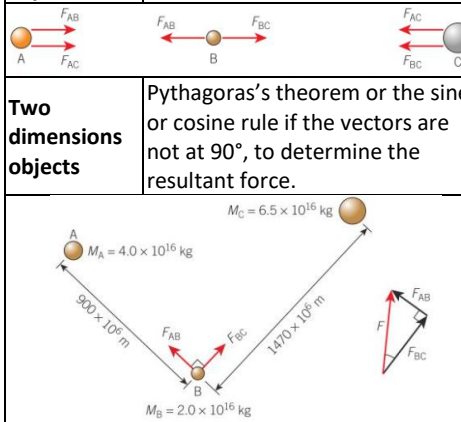
1. Lesson 1 Gravitational fields

Gravitational field	A field created around any object with mass, extending all the way to infinity, but diminishing as the distance from the centre of mass of the objects increases.
Gravitational field strength, g	The gravitational force exerted per unit mass at a point within a gravitational field. Vector quantity, and always points to the centre of mass of the object. $g = \frac{F}{m}$ F – gravitational force (N) m – mass of the object in the gravitational field (kg) g – gravitational field strength (m s^{-2})
$g = a$	Gravitational field strength at a point is the same as the acceleration of free fall of an object at that point.
9.81 N kg^{-1}	The gravitational field strength on the surface of the Earth, at sea level near the equator.
Gravitational field lines or lines of force	Lines of force used to map gravitational field pattern around an object having mass. These lines do not cross, and the arrows on the lines show the direction of the field.
Gravitational force	Is always attractive, the direction of the gravitational field is always towards the centre of mass of the object producing the field.
Gravitational field direction	The direction of the gravitational field is always towards the centre of mass of the object producing the field.

Radial field	A symmetrical field that diminishes with distance from its centre, such as the gravitational field around a spherical mass or the electrical field around a spherical charged object.
Gravitational field stronger	The field lines are closer together.
Gravitational field weaker	The gravitational field strength decreases with distance from the centre of the mass, field lines getting further apart. 
Point mass	A mass with negligible volume. 
Radial field VS point mass	Very similar. We can model a large planet or a star as a point mass, with field lines converging at the centre of mass of the object.
Uniform gravitational field	A gravitational field in which the field lines are parallel and the value for g remains constant. The gravitational field close to the surface of a planet is approximately uniform. 

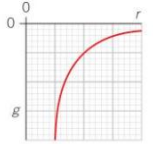
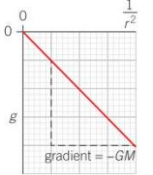
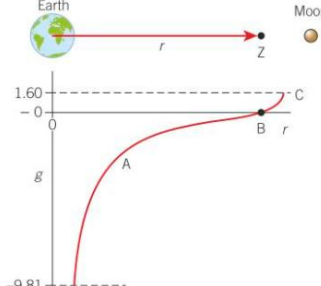
2. Lesson 2 Newton's law of gravitation

Newton's law of gravitation or universal law of gravity	The force between two point masses is directly proportional to the product of the masses and inversely proportional to the square of the separation between them. $F = -\frac{GMm}{r^2}$ M – mass of an object (kg) m – mass of an object (kg) r – distance (m) G – gravitational constant
Minus sign	Gravitational force is attractive.
Gravitational constant	Determined from experiments. $6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$
Two objects	Each object creates its own gravitational field. 
Inverse-square relationship	The attractive force F between objects decreases with distance in an inverse-square relationship. Double the distance and the force between objects will decrease by a factor of four. 

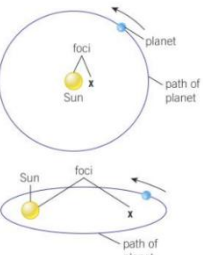
Multiple objects	If several objects are involved, the resultant force can be determined by vector addition.
One dimension objects	The calculation uses addition and subtraction.
Two dimensions objects	Pythagoras's theorem or the sine or cosine rule if the vectors are not at 90° , to determine the resultant force.
	

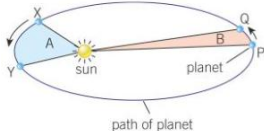
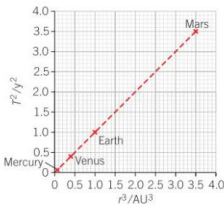
3. Lesson 3 Gravitational field strength for a point mass

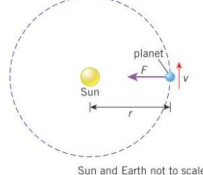
Gravitational field strength g equation	$g = -\frac{GM}{r^2}$ In a radial field the gravitational field strength at a point is: - directly proportional to the mass of the object creating the gravitational field ($g \propto M$) - inversely proportional to the square of the distance from the centre of mass of the object. ($g \propto \frac{1}{r^2}$).
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Negative sign in the equation	$g = -\frac{GM}{r^2}$ The negative sign shows that the gravitational field strength at that point is in the opposite direction to the displacement r from the centre of mass.
Graph of gravitational field strength against distance r	 The values of g are negative. As the distance from the centre of mass increases, g decreases until, at infinity, reaches zero.
Graph of gravitational field strength against $1/r^2$	 The graph of g against $1/r^2$ is a straight line through the origin with a gradient equal to $-GM$.
Uniform gravitational field	The gravitational field strength does not change. Close to the surface of the Earth g is fairly constant. Even at the top of the tallest mountain, the increase in the distance from the centre of mass of the Earth is negligible.
From the Earth to the Moon	

On the surface of the Earth the magnitude of the gravitational field strength is 9.81 N kg^{-1}. The effect of the Moon's gravitational field is quite small. As the spacecraft gets closer to the Moon, the gravitational field strength from the Moon increases. At B on the graph, the gravitational fields of the Earth and the Moon cancel out, meaning there is zero net gravitational field (ignoring the gravitational field from the Sun) at position Z, which is much closer to the Moon than the Earth.
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4. Lesson 4 Kepler's laws	
Johannes Kepler (1571-1630)	German astronomer and mathematician. He published three laws of planetary motion. These laws were based purely on observational data for the then known planets. Kepler had no knowledge of gravitational fields.
Kepler's first law	The orbit of a planet is in ellipse with the Sun at one of the two foci. 
Eccentricity	A measure of the elongation of an ellipse.
Aphelion	The furthest point from the Sun in an orbit.

Perihelion	The closest point to the Sun in an orbit.																																										
Kepler's second law	A line segment joining a planet and the Sun sweeps out equal areas during equal areas of time.  In other words, if the time interval from X to Y is the same as for P and Q the areas A and B must be the same. When a planet is closer to the Sun it moves faster.																																										
Kepler's third law	The square of the orbital period T of a planet is directly proportional to the cube of its average distance r from the Sun. $\frac{T^3}{r^3} = k$ k – constant for the planets orbiting the Sun T – orbital period r – average distance from the Sun 																																										
Astronomical unit (AU)	The mean distance from the Earth to the Sun , i.e. 150 million km or 1.50×10^{11} m.																																										
Data for the six inner planets																																											
<table><tr><th>Planet</th><th>T/y</th><th>T^2/y^2</th><th>r/AU</th><th>r^3/AU^3</th><th>$\frac{T^2}{r^3}/\text{y}^2 \text{AU}^{-3}$</th></tr><tr><td>Mercury</td><td>0.24</td><td>0.058</td><td>0.40</td><td>0.064</td><td>0.91</td></tr><tr><td>Venus</td><td>0.62</td><td>0.38</td><td>0.70</td><td>0.343</td><td>1.11</td></tr><tr><td>Earth</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1.00</td></tr><tr><td>Mars</td><td>1.88</td><td>3.53</td><td>1.50</td><td>3.37</td><td>1.05</td></tr><tr><td>Jupiter</td><td>11.86</td><td>140.6</td><td>5.20</td><td>141</td><td>1.00</td></tr><tr><td>Saturn</td><td>29.46</td><td>867.9</td><td>9.50</td><td>857</td><td>1.01</td></tr></table>	Planet	T/y	T^2/y^2	r/AU	r^3/AU^3	$\frac{T^2}{r^3}/\text{y}^2 \text{AU}^{-3}$	Mercury	0.24	0.058	0.40	0.064	0.91	Venus	0.62	0.38	0.70	0.343	1.11	Earth	1.00	1.00	1.00	1.00	1.00	Mars	1.88	3.53	1.50	3.37	1.05	Jupiter	11.86	140.6	5.20	141	1.00	Saturn	29.46	867.9	9.50	857	1.01	
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	Most planets in the Solar system have almost circular orbits.  Centripetal force on planet = gravitational force on planet $F_{cp} = \frac{mv^2}{r} \quad (1)$ $F = \frac{GMm}{r^2} \quad (2)$ $F_{cp} = F \quad (3)$ Substituting (1) and (2) into (3) $v^2 = \frac{GM}{r} \quad (4)$ The speed v of the planet $v = \frac{2\pi r}{T} \quad (5)$ Substituting (5) into (4) and rearranging $T^2 = \left(\frac{4\pi^2}{GM}\right)r^3$ This equation is a mathematical version of Kepler's third law.
Modelling planetary orbits as circles	
Mathematical version of Kepler's third law	The ratio $\frac{T^2}{r^3}$ is a constant and equal to $\frac{4\pi^2}{GM}$. The gradient of a graph of T^2 against r^3 must be equal to $\frac{4\pi^2}{GM}$.
Kepler's laws application	The laws apply not only to the planets in our Solar system but also to any smaller object in orbit around a larger one. This includes satellites and moons in orbits around planets.

5. Lesson 5 Satellites

Satellite	A body orbiting around the planet.
Gravitational force on any satellite in orbit	$F = \frac{GMm}{r^2}$ m – mass of the satellite (kg) M – mass of the Earth (kg) v – constant speed of the satellite (ms^{-1}) G – gravitational constant r – distance from the satellite to the centre of the Earth (m)
Forces on a satellite	Only one and it is the gravitational attraction between the satellite and the Earth, it is always 'falling towards the Earth'.
Satellite is NOT falling towards the Earth	Because it is travelling so fast, it travels such a great distance that as it falls the Earth curves away beneath it, keeping it at the same height above the surface.
Stable orbit	- exactly the right height - exactly the right speed
Speed of a satellite	$v = \sqrt{\frac{GM}{r}}$ All satellites placed in a given orbit at a given height will be travelling at the same speed, even if their mass varies.
Satellite's constant speed	- no air resistance to slow them down - above the atmosphere
Uses of satellites	
Communications	satellite phones(not mobile phones), TV, some types of satellite radio
Military uses	reconnaissance

Scientific research	Both looking down onto the Earth to monitor crops, pollution, vegetation, and so on, and looking outwards to study the Universe (including several famous examples like the Hubble Space Telescope).
Weather and climate	Predicting and monitoring the weather across the globe and monitoring long-term changes in climate
Global positioning	
Types of orbit	
Polar orbit	The satellite circles the pole. It is useful orbit for mapping and reconnaissance.
Low Earth orbit	Satellites in this kind of orbit take only a short time to orbit the Earth. Less than 2 hours.
Geostationary satellites	Geostationary satellites remain above the same point of the Earth whilst the Earth rotates.

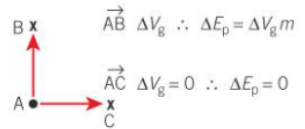
Geostationary satellites must	- be in an orbit above the Earth's equator - rotate in the same direction as the Earth's rotation - have an orbital period of 24 hours.
GPS	Global positioning system – is a series of 32 satellites placed in low Earth orbit.
GPS uses	- satellite navigation - driverless vehicles - geofencing - mining - tectonics
Geofencing	Sending and receiving information when a vehicle, person, or even a pet enters or leaves a specific area.
Mining	Surveying to provide accurate positions both above and underground.
Tectonics	Measuring the movement of seismic faults during even the smallest earthquakes.

6. Lesson 6 Gravitational potential

Gravitational potential V_g	The work done per unit mass to bring an object from infinity to a point in the gravitational field – unit J kg^{-1}. Scalar quantity. All values of gravitational potential are negative except the maximum value.
Maximum gravitational potential	Maximum at infinity and it is equal to zero.

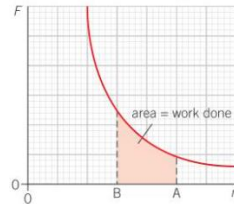
Gravitational potential in radial field	Depends on: - the distance r from the point mass producing the gravitational field to that point - the mass M of the point mass $V_g = -\frac{GM}{r}$ Distance r is measured from the centre of the planet and is greater than or equal to the radius of the planet.
Graph of V_g against r	
Changes in gravitational potential	- moving towards a point mass results in a decrease in gravitational potential - moving away from a point mass (towards infinity) results in an increase in gravitational potential
More masses involved	The total gravitational potential at any point is equal to the algebraic sum of the gravitational potentials from each mass at that point. Since gravitational potential is a scalar quantity.

7. Lesson 7 Gravitational potential energy

Gravitational potential energy	The capacity for doing work as a result of an object's position in a gravitational field. $E = mV_g$ E – gravitational potential energy (J) m – mass of an object within a gravitational field (kg) V_g – gravitational potential (Jkg^{-1})
Changes in gravitational potential energy	For an object with constant mass $\Delta E = m\Delta V_g$
Gravitational potential energy in a uniform gravitational field	 $\vec{AB} \Delta V_g \therefore \Delta E_p = \Delta V_g m$ $\vec{AC} \Delta V_g = 0 \therefore \Delta E_p = 0$ - From A to B: increase in gravitational potential an increase in gravitational potential energy - From A to C: there is no change in gravitational potential, so no change in gravitational potential energy - from B to A: decrease in gravitational potential an decrease in gravitational potential energy
Gravitational potential energy in a radial field	$E = mV_g = -\frac{GMm}{r}$  $V_A = -\frac{GM}{r_A}$ $V_B = -\frac{GM}{r_B}$ $\Delta V_g = V_B - V_A$ $\Delta E_p = \Delta V_g m$

Graphs of force against distance

The gravitational force between objects decreases with distance in an inverse-square relationship.



The area under a force against distance graph is equal to the work done.

The work done can be negative if the object moves from A to B (falls), the object loses gravitational potential energy.

Escape velocity

In order for the projectile to have just enough energy to leave the gravitational field, the loss of kinetic energy must equal to the gain in gravitational potential energy.

$$\frac{1}{2}mv^2 = \frac{GMm}{r}$$

The minimum velocity v for this condition to be met is called the escape velocity.

$$v = \sqrt{\left(\frac{2GM}{r}\right)}$$

The escape velocity on a given planet is therefore the same for all objects regardless of their mass.